Multi-Omics Factor Analysis

Data without Borders - UVRI

Mikhael Dito Manurung, MD/PhD

Leiden University Center for Infectious Diseases (LU-CID)

Leiden University Medical Center

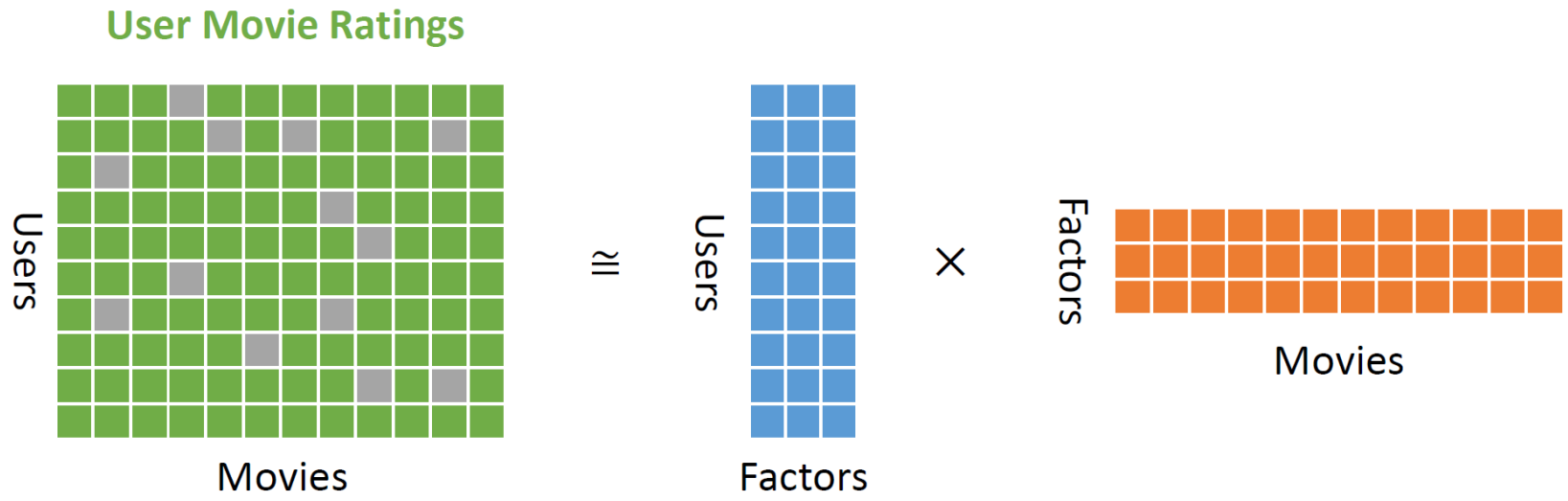
Roadmap

- Why MOFA?
- The MOFA model
- Model parameters and training
- Interpretation and visualization
- Pitfalls and practical tips

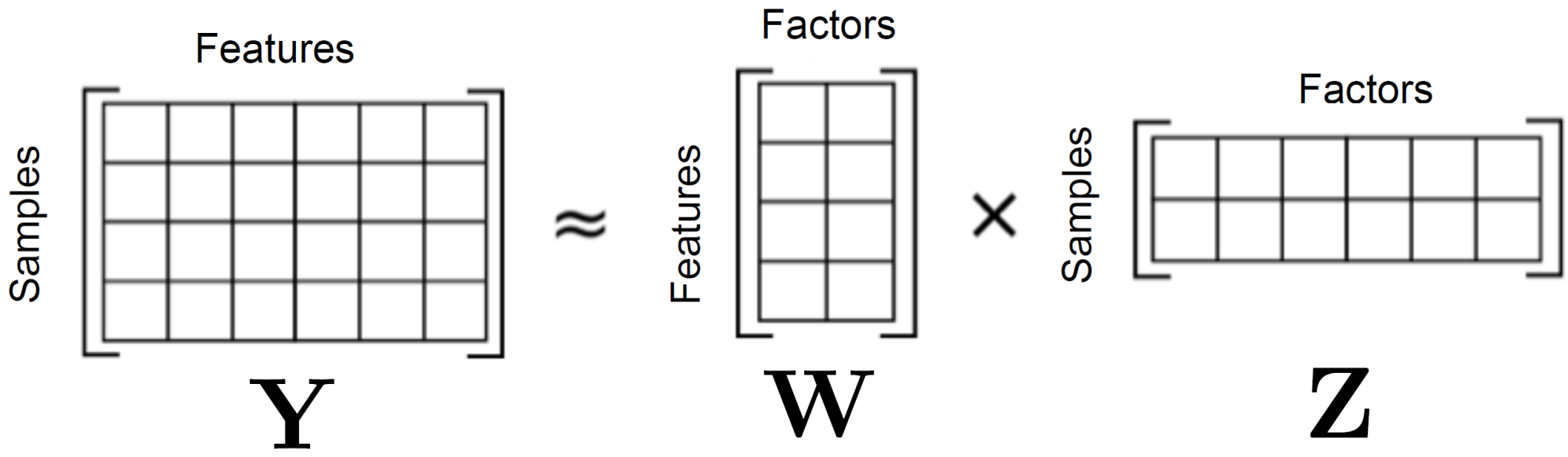
Movie recommendations

- A streaming service has access to the **star ratings** its users have given for different **movies**.
- The service wants to know **how much a user would like another movie** to provide better recommendations

- What could the factors represent in this situation? Do they always represent something “real”?
- What about positive and negative factor scores and loadings?



Matrix factorization



Y is the <u>observed</u> measurements

W is the <u>inferred</u> feature weights

Z is the <u>inferred</u> latent factors

How about PCA?

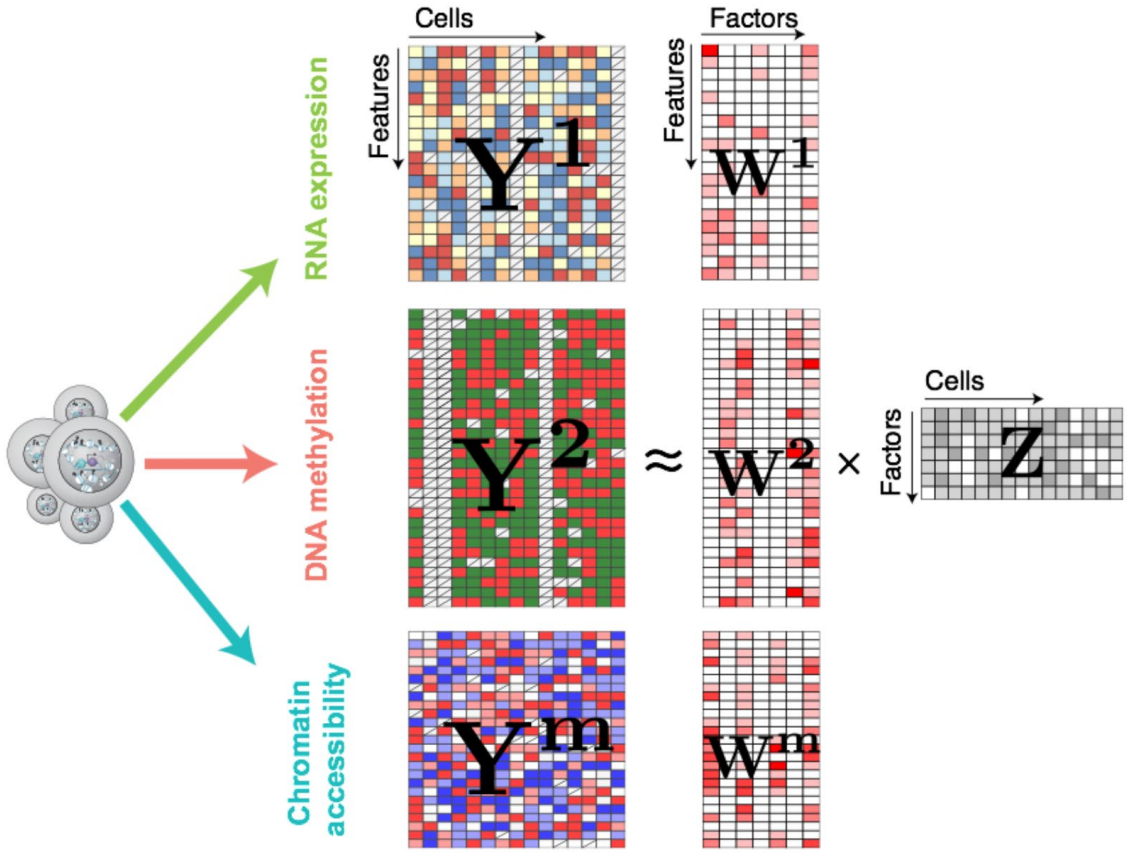
- Does not generalise to an arbitrary number of data modalities
- No natural way to combine different data modalities (binary data with continuous data)
- (Usually) Can’t handle missing values
- Non-sparse loadings: challenges in interpretability and risk of overfitting

Why do we need MOFA?

"What are the main sources of biological variation across multiple omics layers — without knowing the answer in advance?“

- No apriori labels
- Accommodate heterogenous data types
- Accept missing data

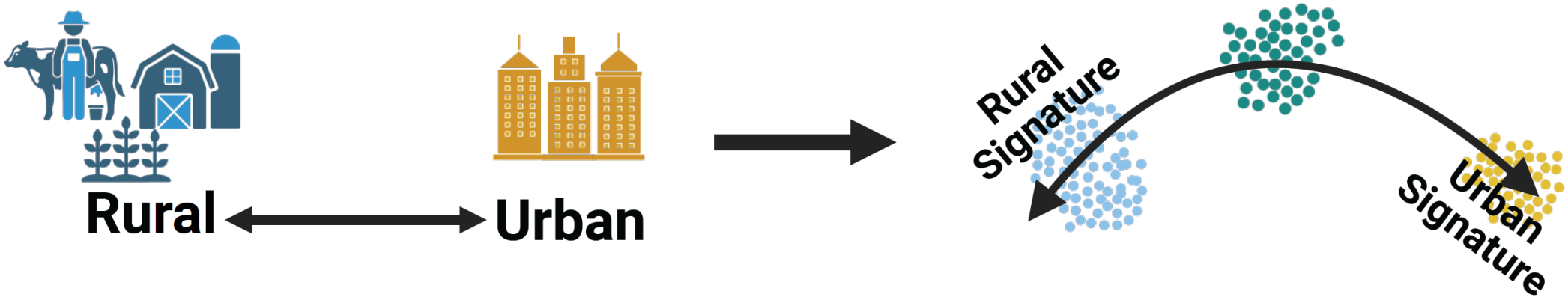
Multi-Omics Factor Analysis (MOFA)

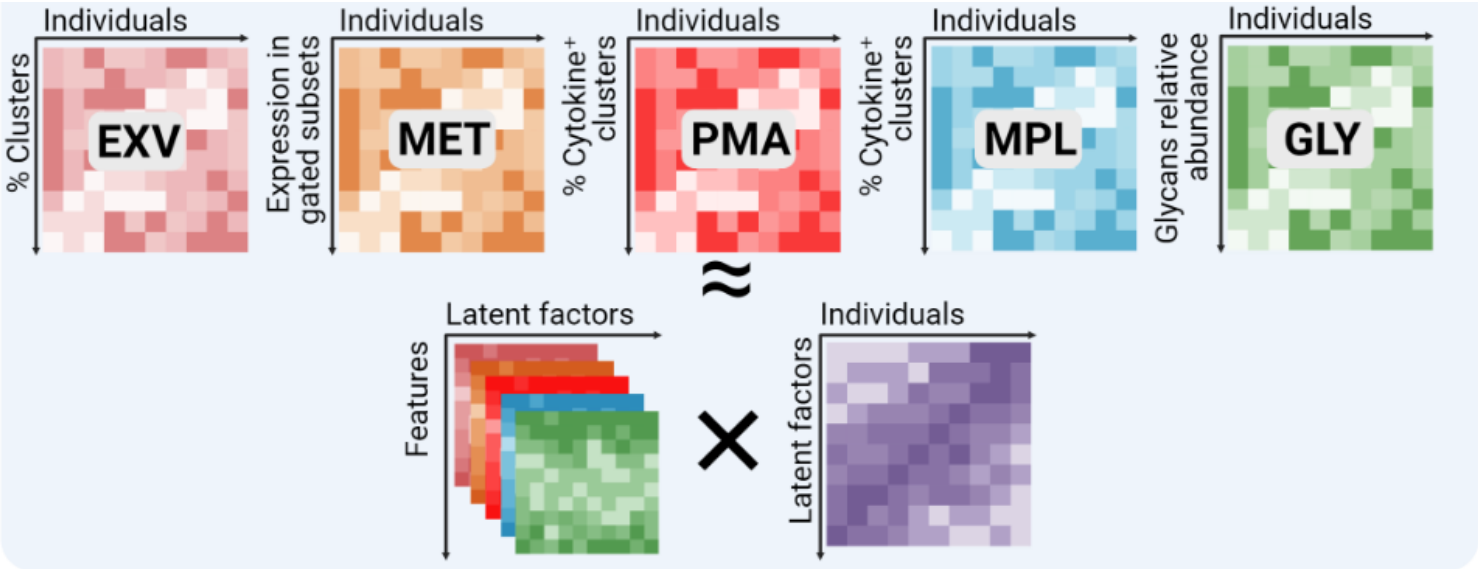


- Generalisation of PCA to multiple data distributions
- Sparsity priors

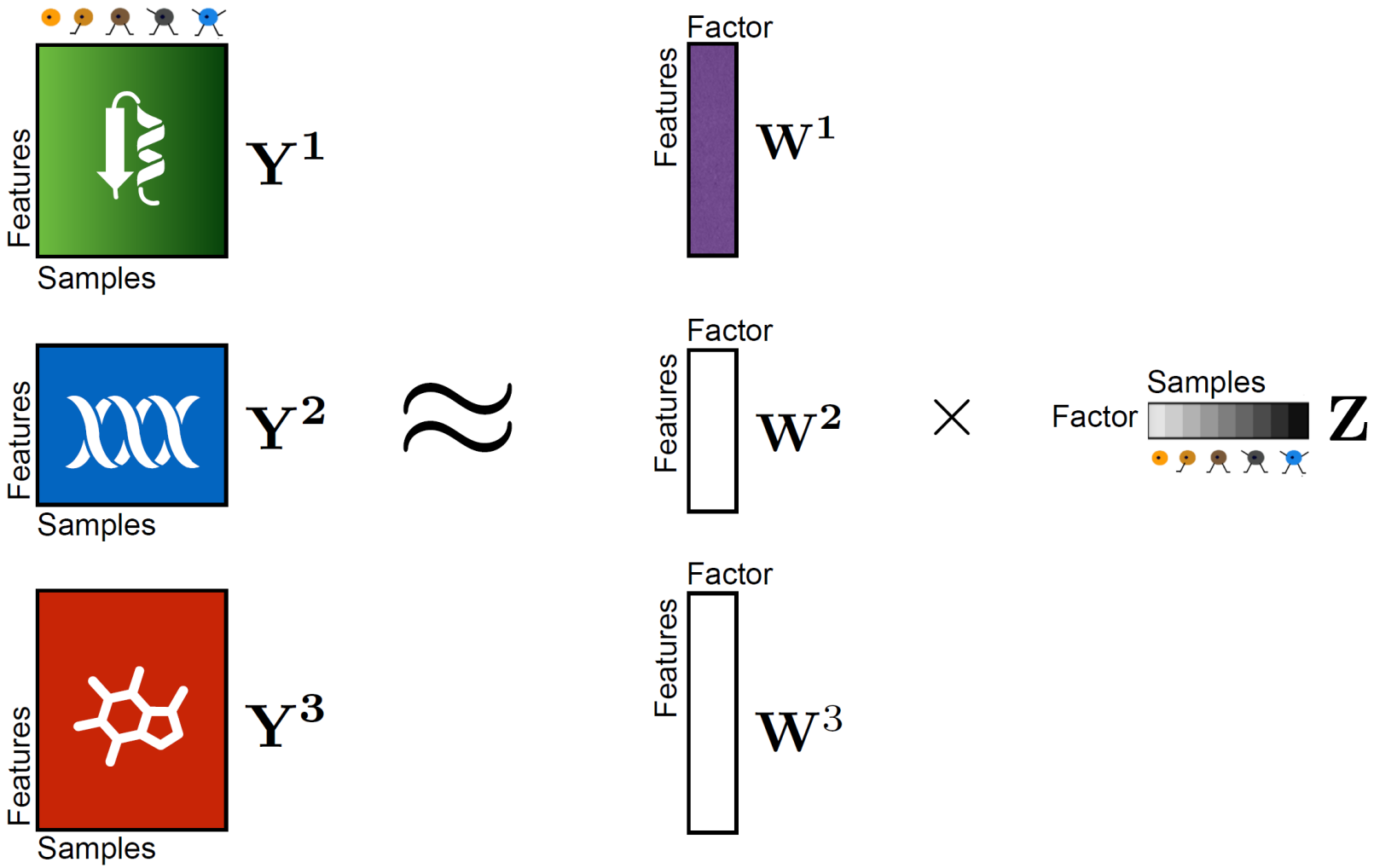
$$\mathbf{Y}^m = \mathbf{Z}\mathbf{W}^{mT}$$

Motivating example

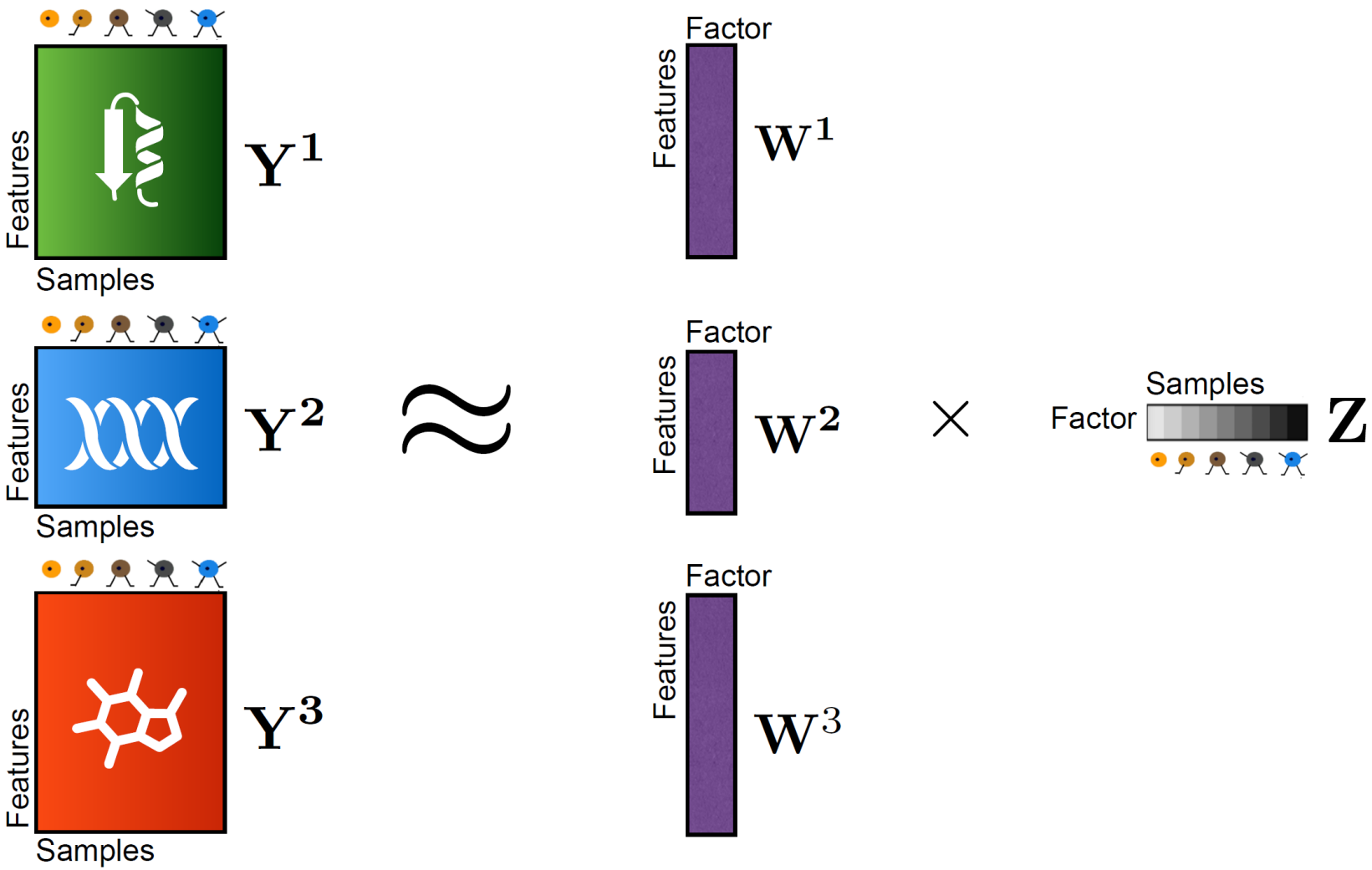




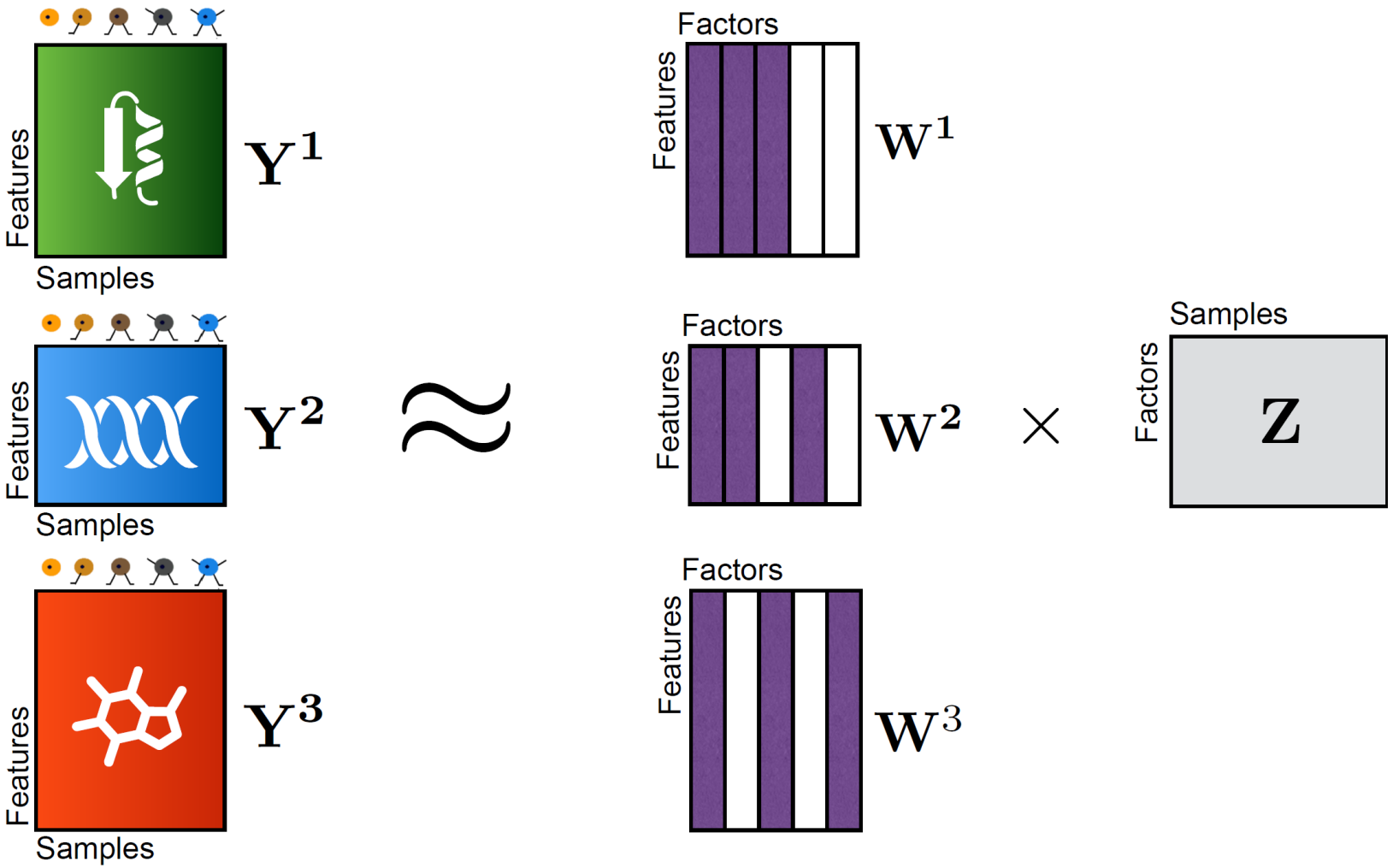
Which modality drives variation? Only one?



Which modality drives variation? All?



A more realistic solution



Model configurations

```
# ── Data options ──────────────────────────────────────────
data_opts <- get_default_data_options(MOFAobject)
# scale_views: scale views to same variance? (FALSE)
# scale_groups: scale groups? (FALSE)

# ── Model options ─────────────────────────────────────────
model_opts <- get_default_model_options(MOFAobject)
model_opts$num_factors <- 15          # upper bound; ARD prunes unused
model_opts$ard_factors <- TRUE

# ── Training options ──────────────────────────────────────
train_opts <- get_default_training_options(MOFAobject)
train_opts$seed            <- 42       # reproducibility
train_opts$convergence_mode <- "fast"  # "fast" / "medium" / "slow"
train_opts$drop_factor_threshold <- 0.01   # min. variance threshold

# ── Prepare and train ─────────────────────────────────────
MOFAobject <- prepare_mofa(MOFAobject,
  data_options     = data_opts,
  model_options    = model_opts,
  training_options = train_opts
)
```

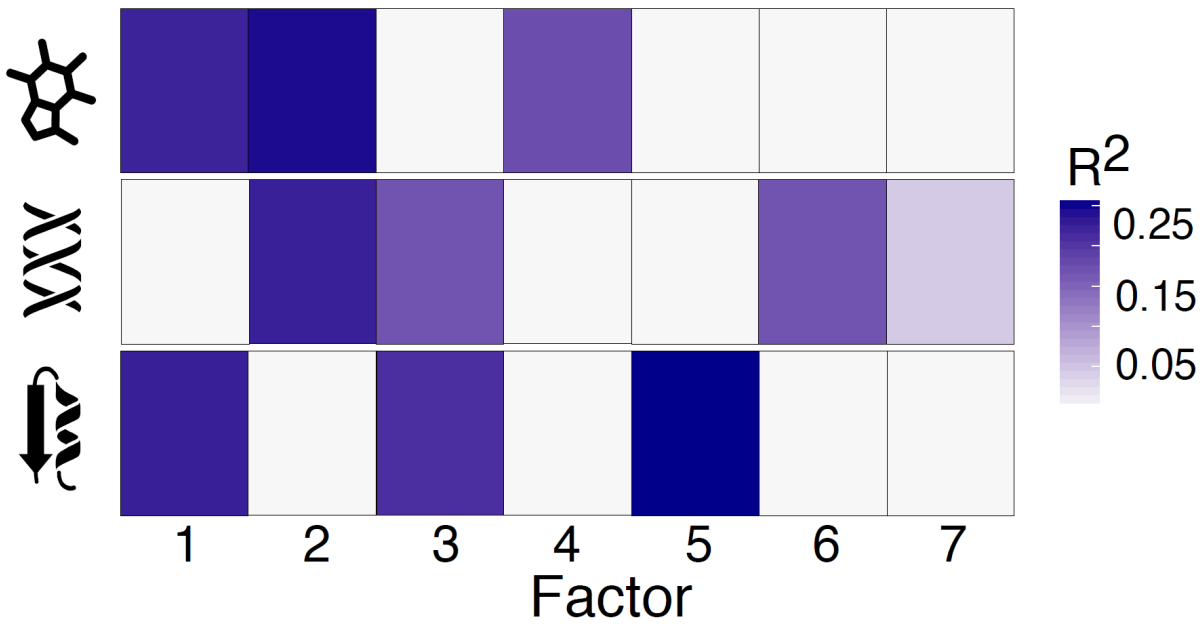
Key options:

num_factors	Upper bound on factors (ARD prunes unused ones)
likelihoods	Per-view: gaussian / bernoulli / poisson
scale_views	Equalise variance across views (FALSE default)
convergence_mode	fast / medium / slow (speed vs accuracy)
seed	Always set for reproducibility

Practical Tip

Set num_factors to a generous upper bound (e.g. 15–25). ARD will automatically prune factors that explain little variance. You don't need to tune this as carefully as ncomp in DIABLO.

Variance decomposition

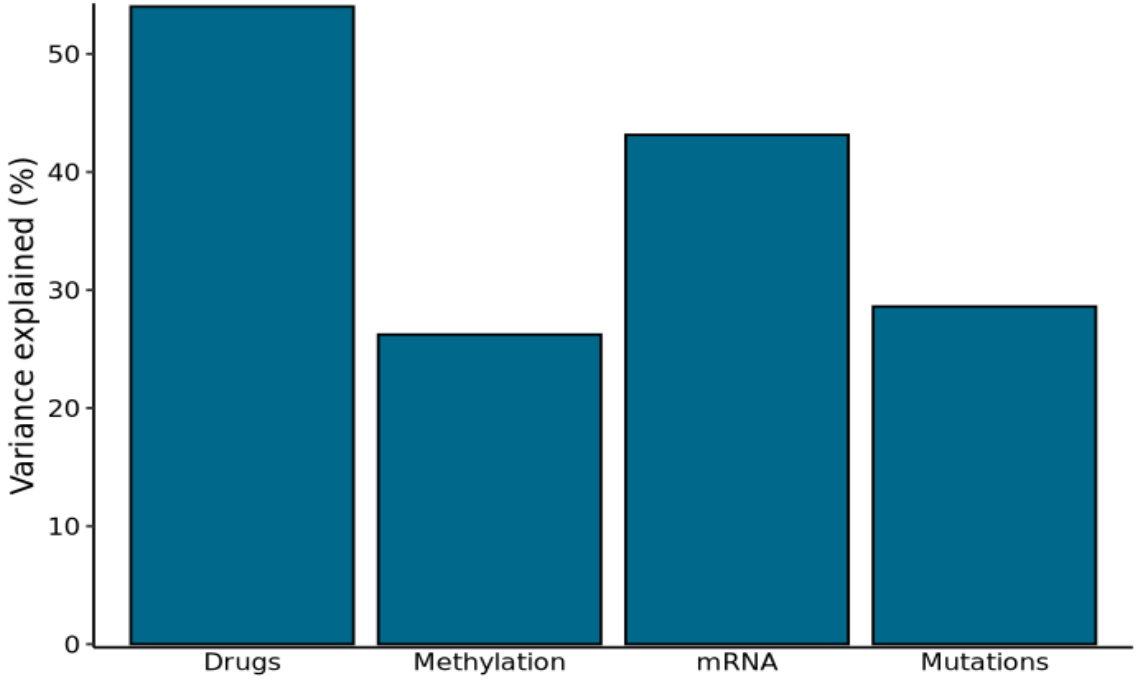


- Rows = factors, Columns = omics views
- Color = % variance explained by that factor in that view
- Dark = high variance explained, white = none
- **Factor active in multiple views → cross-omics biology**
- Factor active in one view only → view-specific signal

```
plot_variance_explained(MOFAobject, max_r2 = 25)
```

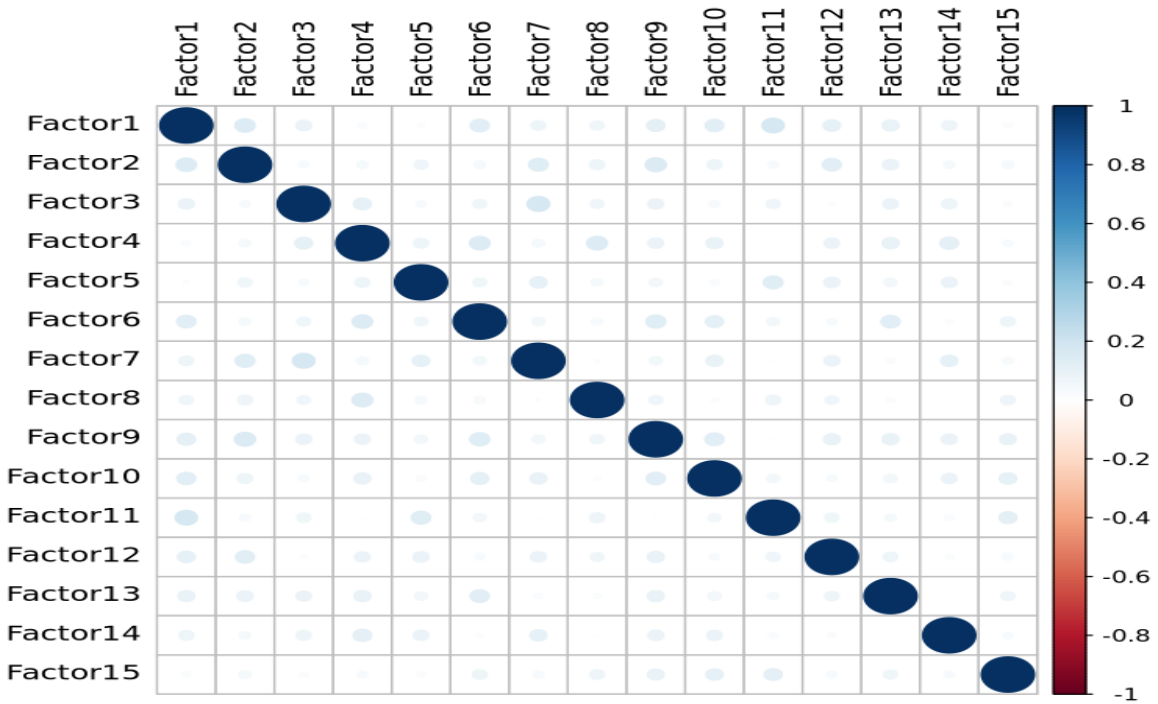
Model Quality Checks

Total variance explained per view:



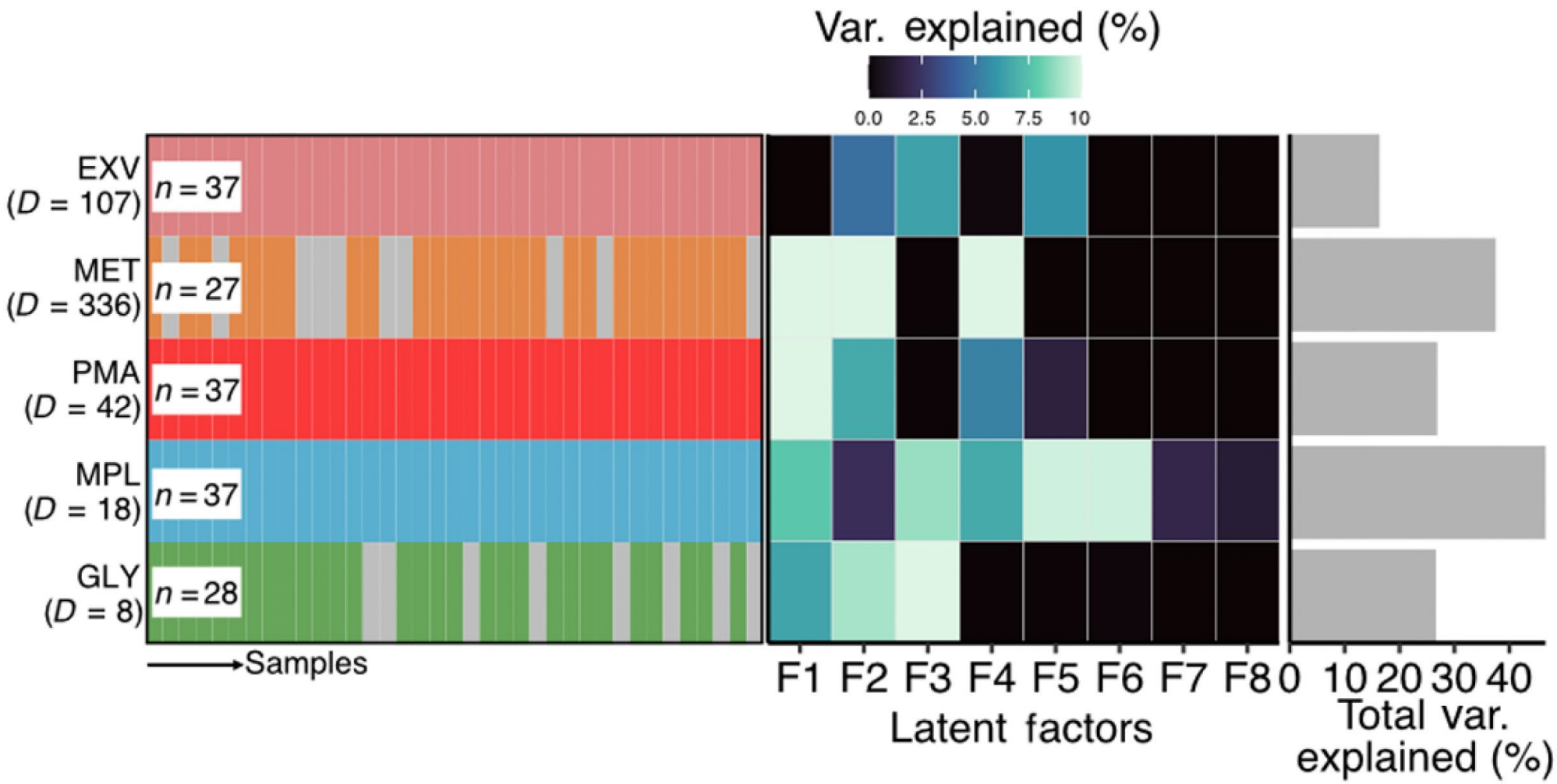

```
plot_variance_explained(MOFAobject,
  plot_total = TRUE)[[2]]
```

Factor correlation matrix:

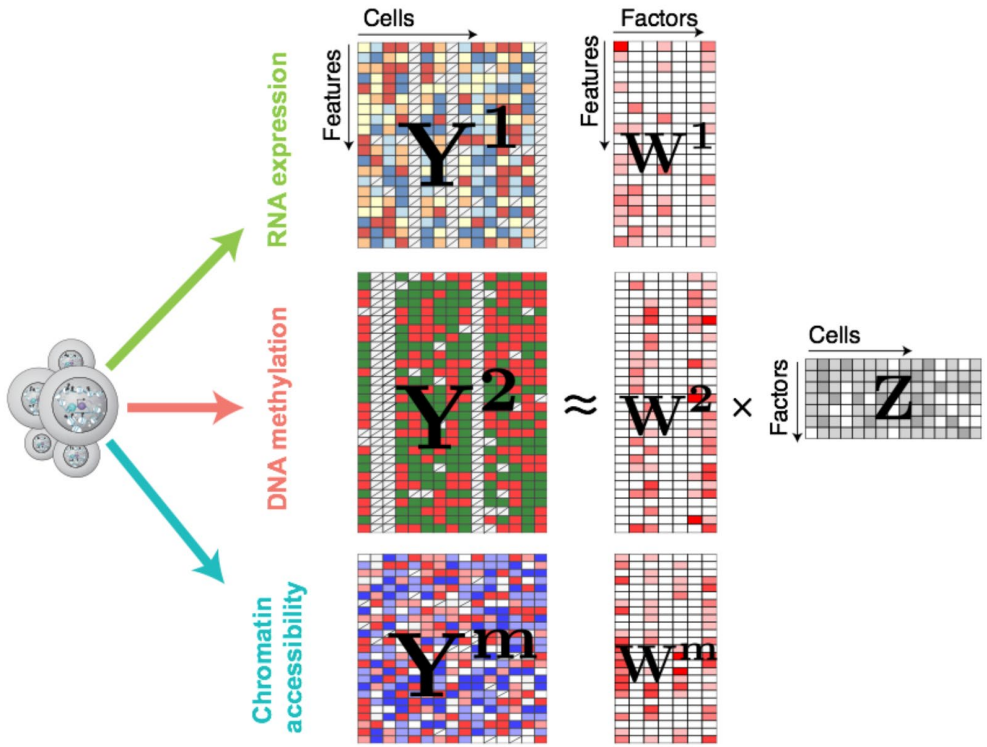



```
plot_factor_cor(MOFAobject)
```

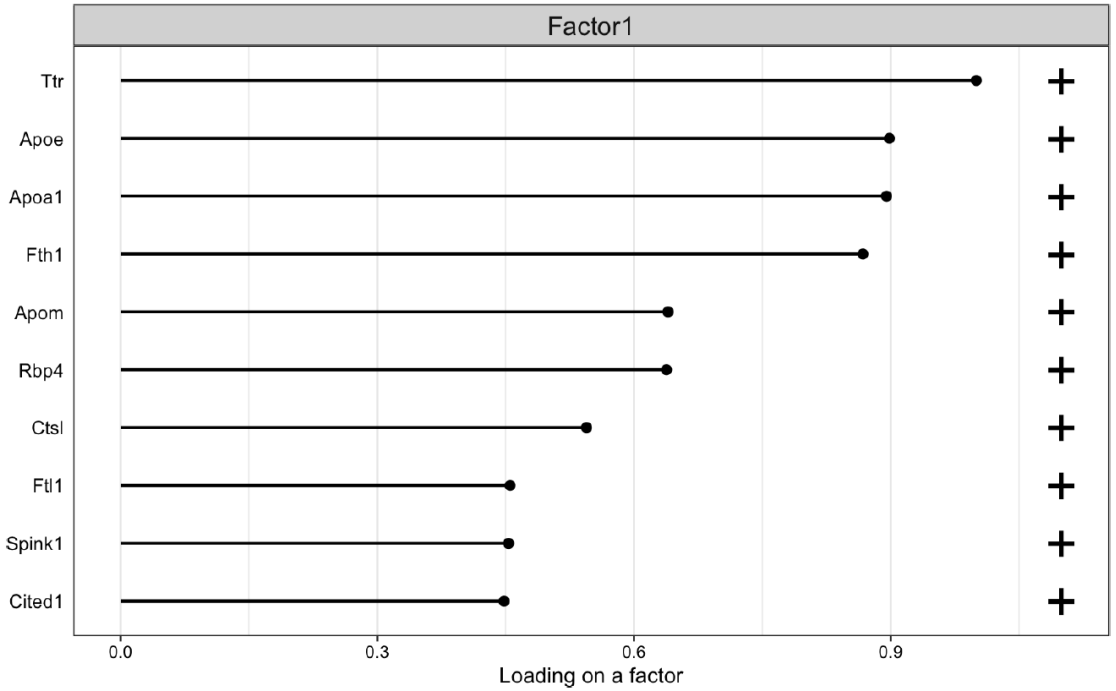
Case study example



Inspection of feature weights

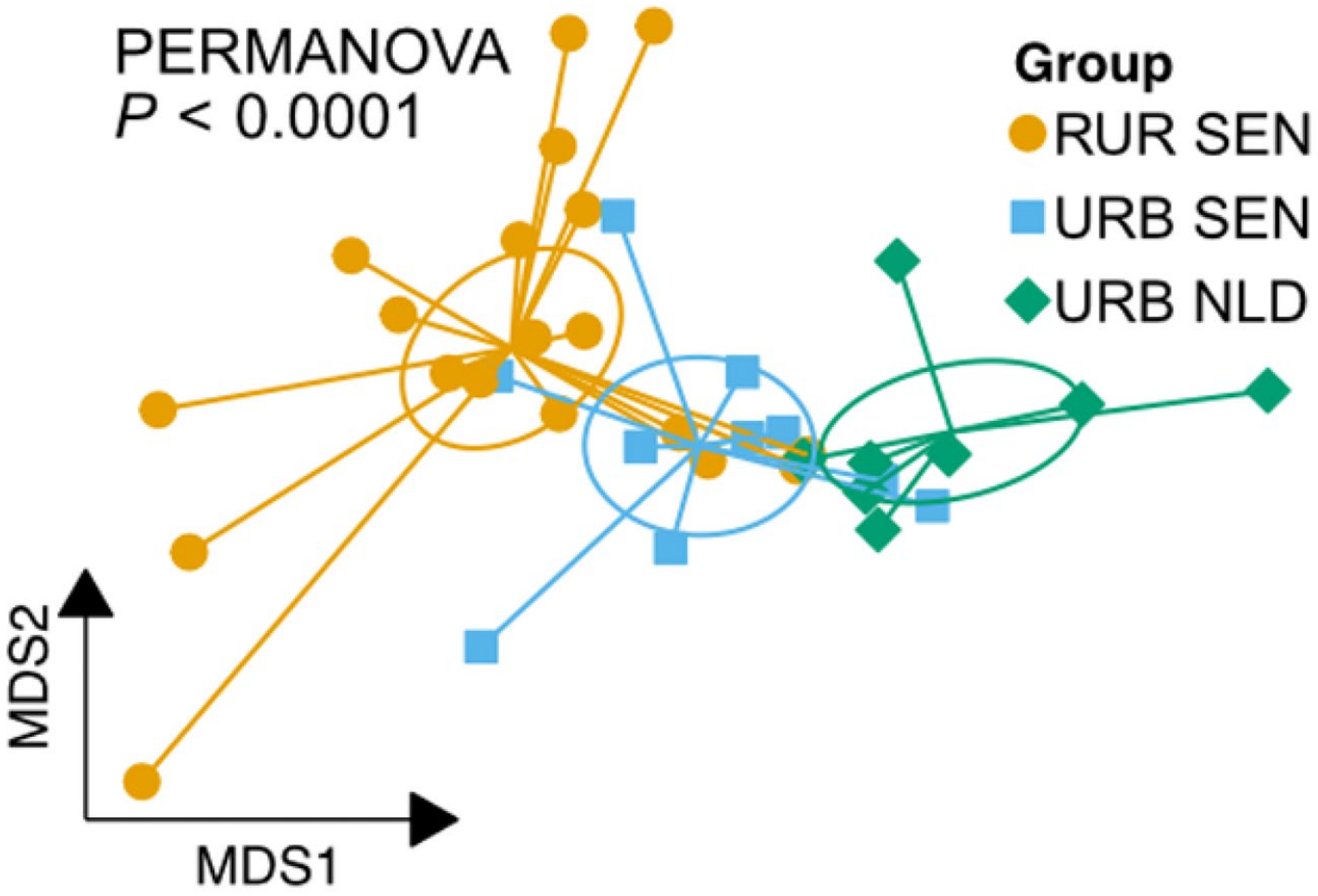


What drives each factor?

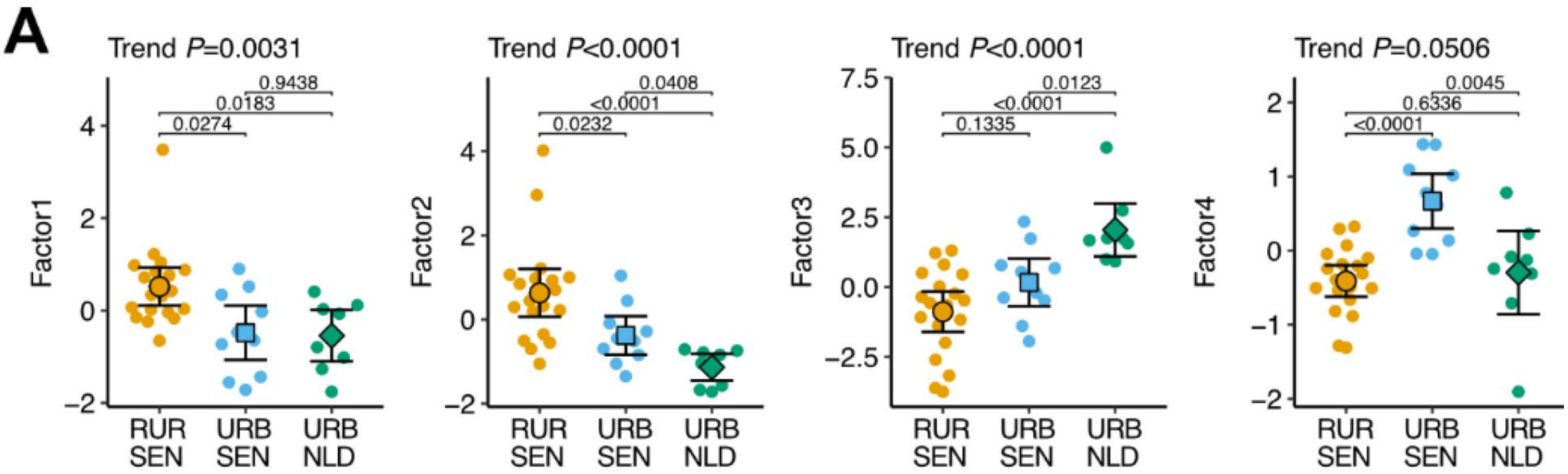


$$\mathbf{Y}^m = \mathbf{Z}\mathbf{W}^{mT}$$

Visualise samples in the latent space



Associating factors with clinical covariates



Imputation and prediction

MOFA+ can reconstruct missing values using the trained model

```
# Impute missing values
MOFAobject <- impute(MOFAobject)

# Access imputed data
get_imputed_data(MOFAobject,
  view = "mRNA")[[1]][1:5, 1:3]
```

- Imputed values stored in @imputed_data slot
- Based on model reconstruction: W(m) · z(n)
- Useful for downstream analyses requiring complete data
- Quality depends on how well the factor explains that view

MOFA vs DIABLO

	MOFA+	DIABLO
Goal	Discover structure (unsupervised)	Discriminate groups (supervised)
Requires Y?	No — no class labels needed	Yes — class labels required
Statistical framework	Bayesian factor analysis	Sparse PLS-DA
Sparsity	ARD (factors) + Spike-slab (weights)	keepX (user-specified feature count)
Data types	Gaussian / Bernoulli / Poisson per view	Continuous only (Gaussian assumed)
Missing data	Handled natively	Must impute first
Multi-group	Yes (MOFA+)	No
Key output	Variance decomposition + factor weights	Latent components + BER classification
Best for	Exploratory analysis, biomarker discovery	Classification, prediction tasks

Common pitfalls

- Sample names must match across all views
- Use appropriate likelihood (binary data → Bernoulli)
- Normalize and batch correct BEFORE MOFA
- Factor signs are arbitrary

Practical tips

- Set generous num_factors (15–25). ARD will prune unused factors automatically.
- Set scale_views = TRUE
- Run model multiple times with different seed!!

Key takeaways

- **MOFA+ is unsupervised — it discovers the main axes of variation across omics without class labels**
- Variance decomposition is the key output — it tells you which factors are biologically important
- ARD + spike-slab priors give automatic, interpretable sparsity — no manual tuning of keepX
- View-specific likelihoods handle heterogeneous data types (continuous, binary, count)
- MOFA+ and DIABLO are complementary: use MOFA+ for discovery, DIABLO for discrimination

Further resources

- MOFA | Multi-Omics Factor Analysis
- The MOFA-FLEX model — mofaflex

Precorded talks

- **MOFA overview**: precorded talk for the VIB workshop (Belgium, 2021), includes the model overview, intuition and a brief discussion of the CLL application.
- **Overview of single-cell multi-omics data integration**: precorded talk for a webinar, includes brief discussion on CITE-seq and 10x Multiome applications.

Getting started using R

- **Training a MOFA model in R**: using simple simulated data
- **Downstream analysis in R**: using simple simulated data
- **Gene set enrichment analysis**: demonstrates how to do gene set enrichment analysis in R.
- **Demonstration of the stochastic inference algorithm**: this is only useful for very large data sets and when having access to GPUs.

Case examples using real data (in R)

- **(authors' favourite) Analysis of chronic lymphocytic leukaemia cohort for personalised medicine**: a bulk multi-omics data set. Figure 2 and 3 of the [MOFA paper (https://www.embopress.org/doi/full/10.15252/msb.20178124#msb178124-fig-0002).
- **Integrative analysis of the Chromium Single Cell Multiome ATAC + Gene Expression assay**: this is the result of a collaboration between the MOFA team and the 10x Genomics R&D team to provide a downstream analysis pipeline for the 10x Multiome kit.
- **Analysis of multi-modal microbiome data**: we demonstrate how to systematically integrate viral, fungal and bacterial sequence data. Manuscript published in mSystems

Practicals!